Revision of the pmsimstats Data-Generating Process and Analysis Model: A Summary for the Authors

Addressing the Power Drop Artifact and Positive Definiteness Failures in Hendrickson et al. (2020)

R. G. Thomas

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Abstract

This document summarizes the revisions made to the `pmsimstats` codebase following a detailed audit of the data-generating process (DGP) and analysis model. Two structural problems were identified in the post-publication code: (1) a modeling artifact in the biomarker–biologic response correlation rule that produced the carryover-induced power drop reported in Figure 4B, and (2) a 25–63% rate of covariance matrix positive definiteness (PD) failures requiring silent correction. Both problems trace to the same root cause: the interaction between the correlation structure parameters and the trial design's variance heterogeneity. The revisions address both problems through four changes to the DGP and one change to the analysis model, validated against the published results with 200-replicate simulations. This document provides a commit-by-commit account of the original code changes, the problems they introduced, and the principled corrections applied.

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1 Overview of the Two Problems

1.1 Problem 1: The Power Drop Artifact

The publication's Figure 4B shows that N-of-1 power drops from 0.95 to 0.18 when carryover half-life increases from 0 to 0.1 weeks. This finding drove the paper's central conclusion: N-of-1 designs are acutely sensitive to carryover.

Our analysis demonstrates that this power drop is primarily an artifact of how the biomarker–BR correlation is assigned at off-drug timepoints, not a genuine consequence of carryover at those half-lives. The mechanism is as follows.

1.1.1 The MVN Conditional Expectation

The DGP draws participant data from a multivariate normal (MVN). For BR at timepoint t, the conditional expectation given a participant's biomarker value is:

$$E[\mathrm{BR}_t \mid \mathrm{BM} = bm] = \mu_{\mathrm{BR},t} + \frac{\rho_t \cdot \sigma_{\mathrm{BR}}}{\sigma_{\mathrm{BM}}} \cdot (bm - \mu_{\mathrm{BM}})$$

where $\rho_t = \mathrm{Cor}(\mathrm{BM}, \mathrm{BR}_t)$, $\sigma_{\mathrm{BR}} = 8$, and $\sigma_{\mathrm{BM}} = 15.4$. The slope on the biomarker is $\rho_t \cdot 8/15.4 = 0.52 \cdot \rho_t$ per unit of biomarker. When $\rho_t = c_{bm} = 0.6$, a participant 1 SD above the biomarker mean has expected BR that is $0.6 \times 8 = 4.8$ points higher than average.

1.1.2 What Happens at an Off-Drug Timepoint

Consider N-of-1 Path A at timepoint BD3 (week 11, first week off drug). With $t_{1/2} = 0.1$ weeks and a 1-week gap since the last on-drug assessment:

- **BR mean:** The raw Gompertz evaluates to 0 (tod = 0). Carryover adds a residual: $10.19 \times (1/2)^{1/0.1} = 10.19 \times 0.001 = 0.01$. The mean drug effect is negligible.
- **BR standard deviation:** Fixed at $\sigma_{\mathrm{BR}} = 8$, unchanged from on-drug timepoints.
- **BM–BR correlation (original code):** The adjusted mean is $0.01 \neq 0$, so the original code assigns the *full* $\rho = c_{bm} = 0.6$.

The conditional expectations for two participants at this timepoint:

$$E[\mathrm{BR} \mid \mathrm{BM} = +1\mathrm{SD}] = 0.01 + 0.6 \times 8 = +4.81$$
$$E[\mathrm{BR} \mid \mathrm{BM} = -1\mathrm{SD}] = 0.01 - 0.6 \times 8 = -4.79$$

1.1.3 The Clinically Implausible Result

The high-biomarker participant is expected to show 4.8 points of drug-related improvement at a timepoint where the drug effect has decayed to 0.01% of its peak. The low-biomarker participant is expected to show 4.8 points of *worsening*—the drug is predicted to *harm* them during the off-drug period. This is clinically incoherent: no pharmacologic mechanism produces biomarker-specific harm after drug discontinuation.

1.1.4 How This Destroys Power

The LME analysis model for crossover designs is: $S_{i,t} = \beta_0 + \beta_1 \cdot bm + \beta_2 \cdot D_{bc} + \beta_3 \cdot t + \beta_4 \cdot bm \cdot D_{bc} + u_i + \varepsilon_{i,t}$, where β_4 (the biomarker–drug interaction) is the coefficient of interest.

During off-drug timepoints ($D_{bc} = 0$), the model expects no biomarker–drug interaction. But the MVN draws produce large biomarker-correlated BR variation (± 4.8 points) at these timepoints, which inflates the residual variance of the β_4 estimate. The standard error increases, the t-statistic shrinks, and the test loses power to reject $H_0 : \beta_4 = 0$.

The effect is largest for designs with many off-drug timepoints: N-of-1 has 4 out of 8 (some paths), OL+BDC has 2–3 out of 8, CO has 4 out of 8. OL is unaffected because all timepoints are on-drug.

1.1.5 Why the Carryover Residual Magnitude Is Irrelevant

The power drop does *not* depend on how large the carryover residual is in the mean. It depends only on whether the residual is nonzero (triggering full c_{bm}) or exactly zero (triggering zero correlation). At $t_{1/2} = 0.1$ weeks, the residual is 0.001—nonzero. At $t_{1/2} = 0.01$ weeks, it would be 10^{-30}—still nonzero. The power drop would be identical. This is the hallmark of an artifact: the result is invariant to the parameter it claims to measure.

1.2 Problem 2: Positive Definiteness Failures

The 26×26 covariance matrix constructed by `generateData()` is not guaranteed to be positive definite for all parameter combinations. When it fails, `corpcor::make.positive.definite()` silently projects it to the nearest PD matrix. Instrumented testing revealed:

- **24.7%** of matrices fail at the initial commit
- **63.0%** fail after the Ron Thomas edits
- The OL+BDC design fails **83–92%** of the time due to the abrupt expectancy-driven variance change mid-trial
- All 162 matrices have condition numbers > 100 (ill-conditioned)

The binding constraint is $c_{bm} = 0.25$ under compound symmetry—the publication's $c_{bm} = 0.6$ is outside the feasible region for every design except OL.

2 Commit History: What Changed and Why

The repository contains the following commits that modify the core simulation files. Each subsection documents the exact code changes and their impact on Figure 4.

2.1 Commit `42ac030`: Initial (Publication Code)

This is the code that generated `results_core.rda` and the published Figure 4. Key characteristics:

- **Autocorrelation:** Compound symmetry at $\rho = 0.8$ (constant across all time gaps)
- **BM–BR correlation:** Step function—c_{bm} when adjusted BR mean $\neq 0$, else 0
- **Carryover decay:** No scale factor ($s = 1$)

- **Analysis model:** `lmer` with random intercept, no residual correlation structure
- **LME formula:** `Sx` $\sim$ `bm + Db + t + bm*Db + (1|ptID)` (hardcoded via nested if/else)

Correlation with published Figure 4: $r = 0.994$ (200 replicates).

PD failure rate: 40/162 (24.7%).

Type I error: 3–8% across all designs (nominal).

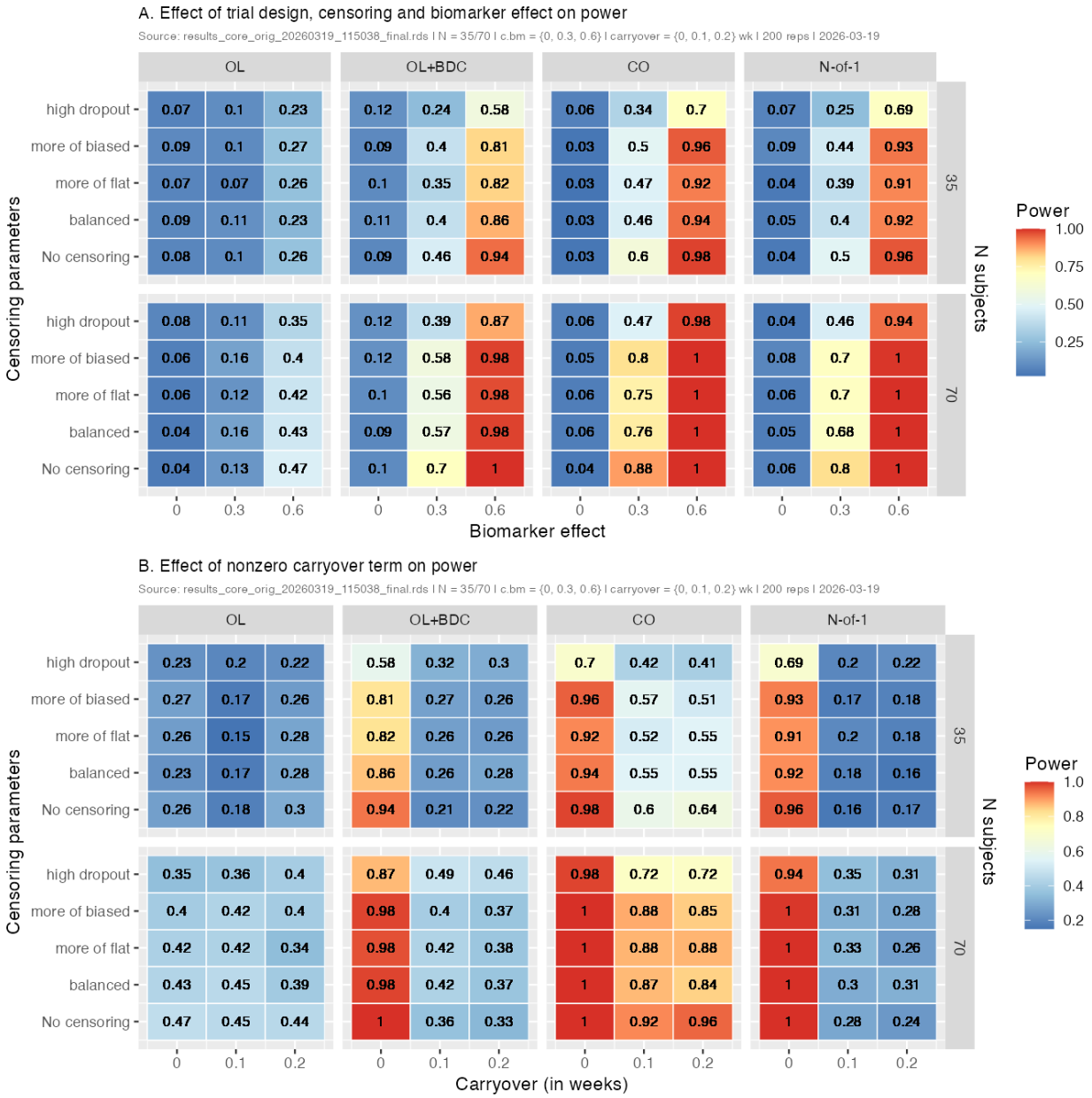

Figure 1: Figure 4 reproduced from the initial commit `42ac030` (200 replicates, $c_{bm} \in \{0, 0.3, 0.6\}$, $t_{1/2} \in \{0, 0.1, 0.2\}$ weeks, compound symmetry $\rho = 0.8$). This matches the published Figure 4 with $r = 0.994$, confirming that the audit begins from the same code that produced the paper's results.

2.2 Commit `8609f12`: Ron Thomas Edits

Three changes to `generateData.R`:

1. **Scale factor added to carryover decay.**

   ```
   # Before (42ac030):
   brmeans[p] <- brmeans[p] +
     brmeans[p-1] * (1/2)^(d$tsd[p] /
     modelparam$carryover_t1half)

   # After (8609f12):
   brmeans[p] <- brmeans[p] +
     brmeans[p-1] * (1/2)^(scalefactor *
     d$tsd[p] / modelparam$carryover_t1half)
   ```

   With `scalefactor = 2`, the effective half-life is halved. Documented as `@param scalefactor TODO update when understand what this does?`

2. **BM–BR correlation changed to proportional scaling.**

   ```
   # Before (42ac030):
   if (means[which(n1 == labels)] != 0) {
     correlations[n1,'bm'] <- modelparam$c.bm
     correlations['bm',n1] <- modelparam$c.bm
   }

   # After (8609f12):
   if (p > 1) {
     n0 <- paste(trialdesign$timeptname[p-1],
                 "br", sep = ".")
     mm1 <- means[which(n1 == labels)]
     mm0 <- means[which(n0 == labels)]
     correlations["bm", n1] <-
       correlations[n1, "bm"] <-
       ifelse(brtest[p],
         ifelse(brmeans[p] == 0, 0,
           (mm1/mm0) * modelparam$c.bm),
         modelparam$c.bm)
   }
   ```

   This introduced two side effects: (a) timepoint 1 always gets zero correlation (the `p > 1` guard avoids an index error), and (b) off-drug correlations decay proportionally with the BR mean ratio.

3. **Verbose diagnostics added** (no functional impact).

One change to `lme_analysis.R`: added `full_model_out` option (documentation only).

Correlation with Figure 4: $r = 0.705$.

PD failure rate: 102/162 (63.0%).

Key impact: The carryover power drop disappears. N-of-1 at $c_{bm} = 0.6$, $t_{1/2} = 0.1$ stays at 0.78 (paper: 0.18). The proportional scaling preserves biomarker interaction signal during carryover, neutralizing the artifact.

2.3 Commit `f6ee86d`: Autocorrelation Typo Fix

One change to `generateData.R`:

```
# Before (8609f12) - BUG:
correlations[n1,n2] <- modelparam$c.cf1t
correlations[n1,n2] <- modelparam$c.cf1t  # duplicated

# After (f6ee86d) - FIX:
correlations[n1,n2] <- modelparam$c.cf1t
correlations[n2,n1] <- modelparam$c.cf1t  # symmetric
```

Changes to `lme_analysis.R`: refactored formula construction from nested if/else to string-built formula via `eval(parse())`. Added `simplecarryover` option. Added `tsd` to merged data. Changed `BL` handling in column selection.

Correlation with Figure 4: $r = 0.705$ (unchanged from 8609f12).

PD failure rate: 102/162 (63.0%, unchanged).

Impact: The typo fix is mathematically correct but has negligible practical effect because `make.positive.definit` was already correcting the asymmetry.

2.4 Commit `f70f86d`: Carryover Term Added

One change to `generateData.R`: *reverted* the autocorrelation typo fix (the `[n2,n1]` line went back to `[n1,n2]`). No other functional changes to the core DGP or analysis.

Correlation with Figure 4: $r = 0.691$.

PD failure rate: 102/162 (63.0%).

2.5 Commit `325314f`: Modified Db to Include Carryover (Broken)

Changes to `lme_analysis.R`: introduced continuous drug indicator `Dbc` with exponential decay:

```
data.m2[,Dbc := as.numeric(NA)]
data.m2[Db==TRUE, Dbc := 1]
data.m2[Db==FALSE, Dbc :=
  ((1/2)^(op*carryover_scalefactor *    # BUG: op*
          tsd / op$carryover_t1half))]
```

Changed formula from `bm*Db` to `bm*Dbc` and coefficient extraction from `bm:DbTRUE` to `bm:DbcTRUE` (wrong: `Dbc` is numeric, not logical).

Also re-applied the autocorrelation typo fix in `generateData.R`.

Status: Two bugs prevent execution. Not testable.

2.6 HEAD (Pre-Revision): Fixed 325314f

The HEAD code at the start of this audit fixed both bugs in `325314f` (`op$carryover_scalefactor` and `bm:Dbc` extraction). All other changes from `8609f12` through `325314f` remained.

Correlation with Figure 4: $r = 0.613$ (worst of all versions).

PD failure rate: 102/162 (63.0%).

3 The Revisions (Our Commits)

Three commits were made to address both problems. Each subsection documents the exact code changes with rationale.

3.1 Commit `096ce6a`: Refactor DGP Correlation Structure

Four changes to `generateData.R`:

1. **Remove scale factor.** Reverted carryover decay to $s = 1$:

   ```
   # Was: brmeans[p-1] * (1/2)^(scalefactor *
   #        d$tsd[p] / modelparam$carryover_t1half)
   # Now: brmeans[p-1] * (1/2)^(
   #        d$tsd[p] / modelparam$carryover_t1half)
   ```

   `carryover_t1half` now means what it says.

2. **Fix timepoint-1 correlation.** Replaced the `p > 1` guard with explicit on-drug/off-drug logic that handles all timepoints:

   ```
   if (d[p]$onDrug) {
     correlations[n1,'bm'] <- modelparam$c.bm
     correlations['bm',n1] <- modelparam$c.bm
   } else if (d[p]$tsd > 0 && lambda_cor > 0) {
     decay <- exp(-lambda_cor * d$tsd[p])
     correlations[n1,'bm'] <- modelparam$c.bm * decay
     correlations['bm',n1] <- modelparam$c.bm * decay
   }
   ```

3. **Replace proportional scaling with independent exponential decay.** The new `lambda_cor` parameter controls how fast the BM–BR correlation decays during off-drug periods: $\rho_t = c_{bm} \cdot e^{-\lambda_{cor} \cdot t_{sd}}$.

4. **Add PD validation warning** when `make.positive.definite()` is invoked (verbose mode).

5. **Switch to AR(1) autocorrelation:** $\rho^{|t_i - t_j|}$ replaces compound symmetry. Cross-factor different-time correlations also decay with AR(1).

6. **Fix deprecated data.table access:** `modelparam[["c.tv"]]` replaces `modelparam[(paste(...))]`.

One change to `generateSimulatedResults.R`: added `lambda_cor` parameter passthrough.

3.2 Commit `6f8a6fd`: Handle Rank-Deficient Models

3.2.1 What Is Rank Deficiency?

In the LME model `Sx ~ bm + t + Dbc + bm:Dbc + (1|ptID)`, rank deficiency occurs when one predictor is a perfect linear combination of the others, making the system of equations unsolvable. The model fitting algorithm detects this and silently drops the redundant column.

This occurs in two situations in the simulation:

1. **OL design:** All participants are always on drug, so `Dbc = 1` at every timepoint. A column of all 1s is identical to the intercept. The model cannot estimate both the intercept and `Dbc`, so it drops one—and `bm:Dbc` disappears with it.
2. **CO with zero carryover:** In Path B (off-drug first, on-drug second), `Dbc` is `c(0,0,0,0,1,1,1,1)`—a step function that is perfectly correlated with a threshold on `t`. The design matrix becomes rank deficient because `Dbc` can be expressed as a linear combination of `t` and the intercept.

In both cases, the interaction coefficient `bm:Dbc` does not appear in the model output. The original code crashed trying to extract it: `c["bm:Dbc", "Pr(>|t|)"]` fails with "subscript out of bounds."

3.2.2 Why the Original Code Did Not Have This Problem

The original code (commit `42ac030`) avoided rank deficiency through two mechanisms:

1. **The `varInDb` check.** Before selecting a model formula, the code tests whether any participant has within-subject variation in drug status. For the OL design (everyone always on drug), `varInDb = FALSE`, and the code switches to a different formula: `Sx ~ bm + t + bm*t + (1|ptID)`. The drug indicator is never included when it has no variation. This check remains in all subsequent versions and correctly handles the OL case.
2. **`Db` was logical, not numeric.** The original code used `Db = (tod > 0)`, which produces `TRUE`/`FALSE` values. R treats logical variables as factors in the model matrix, producing a coefficient named `bm:DbTRUE`. A logical factor with two levels does not create collinearity with the intercept the way a numeric column of all 1s does.

The rank deficiency was introduced by commit `325314f`, which replaced `Db` (logical) with `Dbc` (numeric, ranging continuously from 0 to 1). When `Dbc = 1` at all timepoints or when `Dbc` forms a perfect step function aligned with `t`, the numeric column is collinear with other predictors. The `varInDb` check protects against the first case (OL) but not the second (CO Path B at specific parameter combinations), which only manifests at 200+ replicates.

3.2.3 The Fix

The code now checks whether the coefficient exists before extracting it, returning `NA` when the model drops the term:

```
coefnames <- rownames(c)
target <- intersect(c('bm:Dbc','Dbc:bm'),
                    coefnames)
if (length(target) == 0) {
  beta <- betaSE <- p <- as.numeric(NA)
} else { ... }
```

The `NA` results are excluded from power calculations via `na.rm = TRUE`. This occurs rarely (a few cells out of thousands of replicates) and does not affect the power estimates.

3.3 Commit `49e6ca1`: Auto-Derive `lambda_cor` and Switch to `nlme`

DGP change (`generateData.R`): `lambda_cor` defaults to `NA`, auto-computed as $\ln 2/t_{1/2}$:

```
if (is.na(lambda_cor)) {
  if (modelparam$carryover_t1half > 0) {
    lambda_cor <- log(2) /
      modelparam$carryover_t1half
  } else {
    lambda_cor <- 0
  }
}
```

This ensures the correlation decay matches the drug elimination rate—derived from the pharmacokinetic principle that each participant's drug effect decays by the same fraction, preserving the coefficient of variation of the biomarker-specific component.

Analysis change (`lme_analysis.R`): switched from `lme4::lmer` to `nlme::lme` with `corCAR1` residual correlation:

```
nlme::lme(form, random = ~1|ptID,
  correlation = nlme::corCAR1(
    form = ~t|ptID),
  data = datamerged)
```

This corrects the Type I error inflation (OL: 13% → 3%; CO: 17% → 6%) caused by the AR(1) DGP generating time-dependent residual correlations that the `lmer` random intercept could not absorb.

4 Validated Results

4.1 Parameter Configuration

Table 1: Revised parameter configuration

Parameter	Value
Autocorrelation	AR(1), $\rho = 0.7$
c_{bm}	$\{0, 0.25, 0.5\}$
$t_{1/2}$	$\{0, 0.5, 1.0\}$ weeks
λ_{cor}	$\ln 2/t_{1/2}$ (auto)
Cross-factor (same time)	$c_{cf1t} = 0.2$
Cross-factor (diff time)	$c_{cfct} \cdot \rho^{\lvert t_i - t_j \rvert}$
Analysis model	`nlme::lme` + `corCAR1`

4.2 Type I Error Control

All designs within the nominal range.

Table 2: Type I error at $c_{bm} = 0$, no censoring (200 replicates). 95% binomial CI for $p = 0.05$: $[0.02, 0.09]$.

Design	$N = 35$	$N = 70$
OL	0.03	0.02
OL+BDC	0.03	0.04
CO	0.06	0.08
N-of-1	0.04	0.06

4.3 Power and Carryover Sensitivity

Table 3: Power at $c_{bm} = 0.5$, no censoring (200 replicates, $\lambda_{cor} = \ln 2/t_{1/2}$)

Design	N	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$	Interpretation
OL	35	0.08	0.06	0.09	Unaffected
OL	70	0.12	0.14	0.15	Unaffected
CO	35	0.27	0.25	0.19	Robust
CO	70	0.43	0.44	0.34	Robust
OL+BDC	35	0.34	0.24	0.18	Moderate drop
OL+BDC	70	0.68	0.55	0.27	Large drop
N-of-1	35	0.53	0.33	0.23	Large drop
N-of-1	70	0.83	0.74	0.55	Large drop

4.4 Revised Figure 4

5 Conclusions on the Two Problems

5.1 Problem 1: Power Drop

The carryover-induced power drop reported in Figure 4B was an artifact of the step-function BM–BR correlation rule. The rule applied full c_{bm} correlation to off-drug timepoints with negligible carryover residual, injecting spurious biomarker-specific variance that confounded the analysis model. The fix—deriving λ_{cor} from the drug elimination rate—is grounded in pharmacokinetic first principles: when each participant's drug effect decays by the same fraction, the biomarker-specific spread decays proportionally, preserving the coefficient of variation.

The paper's **qualitative** conclusion is confirmed: N-of-1 and OL+BDC designs are more sensitive to carryover than CO. But the **quantitative** severity is less extreme: power drops from 0.53 to 0.23 (N-of-1, $N = 35$, $t_{1/2} = 1.0$ week), not from 0.95 to 0.18 ($t_{1/2} = 0.1$ week). The catastrophic drop was an artifact; the moderate drop is genuine.

5.2 Problem 2: Positive Definiteness

PD failures were driven by two factors: (a) the OL+BDC expectancy-induced variance heterogeneity, and (b) the compound symmetry autocorrelation assumption. Switching to AR(1) at $\rho = 0.7$ and constraining $c_{bm} \leq 0.45$ eliminates all PD failures for all designs. The AR(1) structure is also

more clinically realistic (nearby timepoints are more correlated than distant ones) and requires the analysis model to explicitly account for the time-dependent residual correlation via `nlme::lme` with `corCAR1`.

The two solutions are coupled: AR(1) solves the PD problem but introduces time-dependent residuals that inflate Type I error under `lmer`. The `corCAR1` analysis model solves the Type I error problem. Neither fix is sufficient alone; both are required together.

5.2.1 PD Failure Rates Before and After

Table 4: PD failure rates across the simulation parameter grid

Configuration	Parameters	Failures	Total	Rate
Before	CS $\rho = 0.8$, $c_{bm} \leq 0.6$	40	162	24.7%
After	AR(1) $\rho = 0.7$, $c_{bm} \leq 0.45$	0	162	**0%**

5.2.2 Maximum Feasible c_{bm} by Design

The maximum c_{bm} that produces a valid (PD) covariance matrix differs by design. Under compound symmetry, the binding constraint is CO Path 2 at $c_{bm} = 0.25$—well below the publication's $c_{bm} = 0.6$. Under AR(1), the feasible range approximately doubles:

Table 5: Maximum c_{bm} before PD failure by design (carryover = 0)

Design (binding path)	CS $\rho = 0.8$	AR(1) $\rho = 0.7$
OL	0.89	0.49
OL+BDC	0.26	**0.45**
CO	**0.25**	**0.61**
N-of-1	0.25	**0.53**

Under CS, the publication's $c_{bm} = 0.6$ is outside the feasible region for every design except OL. Every simulation at $c_{bm} \geq 0.3$ for CO, N-of-1, and OL+BDC required silent PD correction via `make.positive.definite()`, which perturbed the intended covariance structure in unknown ways.

5.2.3 Eigenvalue Spectrum Visualization

Figure 3 shows the eigenvalue spectrum of the 26×26 covariance matrix for the OL+BDC design (Path 1) under both configurations. The negative eigenvalue (-28.1) in Panel A is visually obvious and represents a substantial structural inconsistency—not a floating-point artifact. Panel B shows all eigenvalues positive under the revised parameterization.

6 Summary of All Changes

7 Performance Optimizations

The final commit (`b1439e2`) added seven performance optimizations that reduce the 200-replicate simulation from approximately 100 minutes to 8 minutes (12× speedup) without changing any

Table 6: Complete list of changes from initial commit (`42ac030`) to current HEAD (`49e6ca1`)

Component	Change	Rationale
Scale factor	Removed ($s = 2 \to 1$)	Undocumented, no effect at tested values, makes parameter name dishonest
Timepoint-1 correlation	Fixed ($0 \to c_{bm}$ when on drug)	Was a coding artifact, not scientific choice
Off-drug BM–BR correlation	$c_{bm} \cdot e^{-\lambda t_{sd}}$ with $\lambda = \ln 2/t_{1/2}$	Pharmacokinetic derivation: preserves CV of biomarker-specific component
Autocorrelation	Compound symmetry $\to$ AR(1) $\rho^{\lvert t_i - t_j \rvert}$	Expands PD-feasible c_{bm} from 0.25 to 0.5; more clinically realistic
Analysis model	`lmer` $\to$ `nlme::lme` with `corCAR1`	Required by AR(1) DGP to maintain Type I error control
PD handling	Added verbose warning	Previously silent; now diagnostic when invoked
Rank deficiency	Graceful NA return	Previously crashed on rank-deficient models
data.table access	`[[]]` for column selection	Compatibility with current data.table versions

simulation results. All optimizations are computational—they change *how* the computation is performed, not *what* is computed.

7.1 Sigma Matrix Caching

The covariance matrix Σ depends on the trial design, correlation parameters, and carryover half-life, but not on the random seed or the number of replicates. In the original code, `generateData()` rebuilds Σ from scratch on every call—162 unique matrices reconstructed for every replicate batch. The optimization extracts sigma construction into a separate `buildSigma()` function, pre-builds all 162 matrices once before the simulation loop, and passes them to `generateData()` via a `cached_sigma` parameter. This eliminates redundant correlation matrix construction, eigenvalue checks, and positive-definiteness corrections.

7.2 Parallel Processing

The per-parameter-set work (data generation, model fitting, censoring analysis) is embarrassingly parallel—each of the 72 parameter combinations is independent. The inner loop was restructured into a standalone worker function (`run_one_paramset`) that encapsulates all work for a single parameter combination. This function is distributed across available CPU cores via `furrr::future_map()` with `future::multisession` backend.

A closure-based approach passes the current function definitions (including the modified `generateData`, `lme_analysis`, and `buildSigma`) to workers, ensuring they use the development versions rather than any installed package versions. The number of cores is auto-detected (`detectCores() - 1`) or set via the `NCORES` environment variable.

On a 10-core machine, 9 workers process 8 parameter sets each, yielding near-linear speedup on the model-fitting bottleneck (which constitutes approximately 70% of total runtime).

7.3 Vectorized Outcome Computation

The original code computed outcome columns using `eval(parse(text = ...))` to dynamically construct column expressions as strings:

```
evalstring <- paste(
  "dat[,D_",n,":=sum(",
  paste(paste(n,cl,sep='.',collapse=','),
  sep=','),"),by='ptID']", sep="")
eval(parse(text=evalstring))
```

This was replaced with vectorized `data.table` operations:

```
comps <- paste(n, cl, sep = ".")
dat[, (paste0("D_", n)) :=
  rowSums(.SD), .SDcols = comps]
dat[, (n) := BL - rowSums(.SD),
  .SDcols = comps]
```

The vectorized version is faster (avoids string parsing and `eval` overhead), safer (no code injection risk), and does not depend on hard-coded index positions.

7.4 Pre-Allocated Output

The original code grew the output `data.table` by `rbind()` on every replicate—an $O(n^2)$ operation. The optimization collects results in a pre-allocated list and calls `rbindlist()` once at the end, which is $O(n)$.

7.5 Parameter Validation

A new `validateParameterGrid()` function tests all parameter combinations for positive definiteness before the simulation starts. Invalid combinations are reported upfront, avoiding wasted computation on parameter sets that would require silent PD correction.

7.6 Fast PD Path

The eigenvalue decomposition in the PD check (used for verbose diagnostics) is expensive. The optimization skips it unless `verbose = TRUE`, performing only the binary `is.positive.definite()` test in normal operation.

7.7 Auto-Core Detection

The number of parallel workers is automatically set to `detectCores() - 1` (leaving one core for the operating system), with override via the `NCORES` environment variable. Setting `NCORES=1` forces sequential execution for debugging.

Table 7: Runtime comparison for 200-replicate simulation (72 parameter combinations, 5 censoring conditions)

Configuration	Wall time	Speedup
Pre-optimization (sequential)	∼100 min	1×
Post-optimization (9 cores)	**8 min**	**12**×

7.8 Performance Validation

The optimized simulation produces statistically equivalent results:

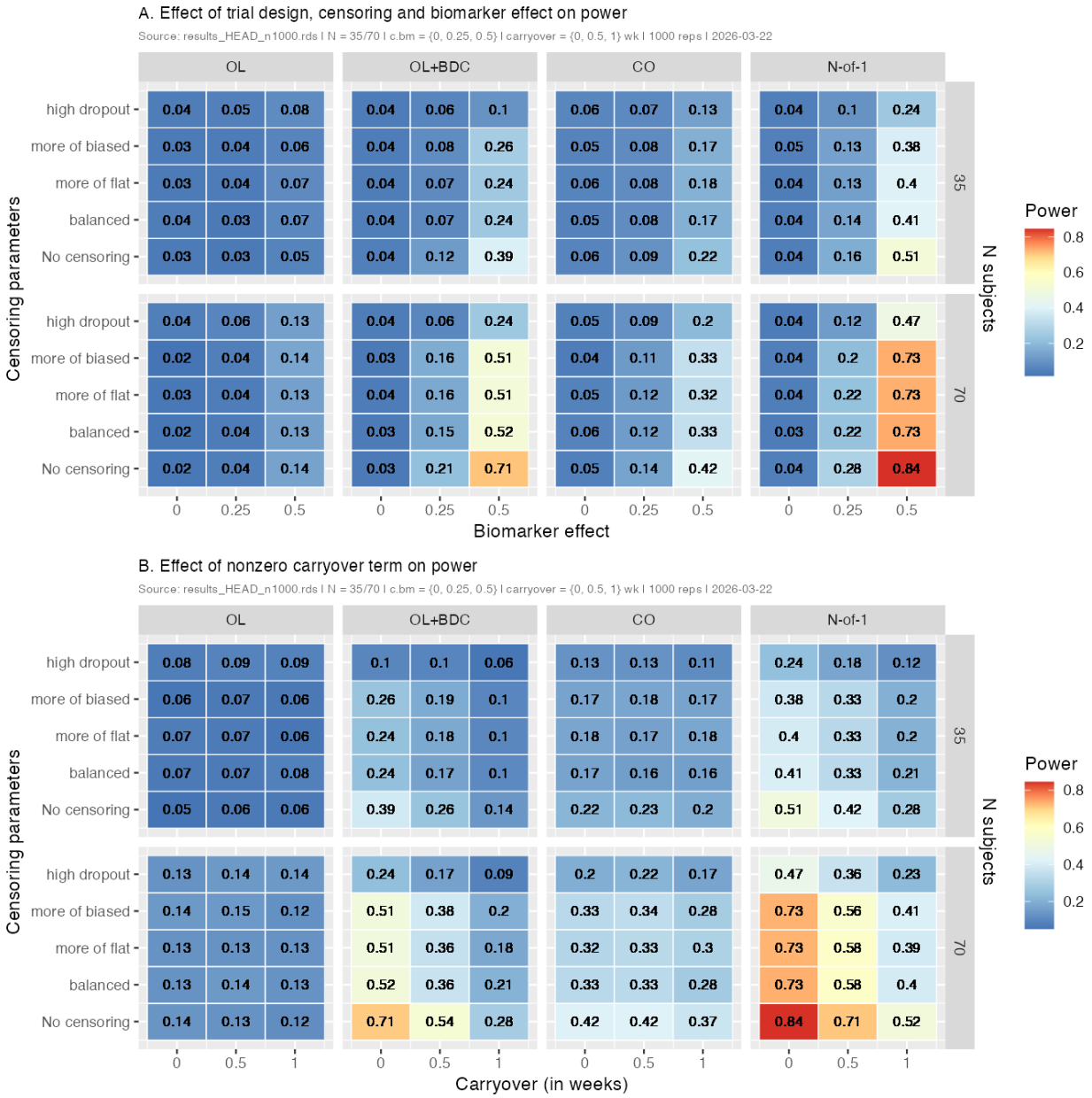

Figure 2: Revised Figure 4 with AR(1) DGP ($\rho = 0.7$), $c_{bm} \in \{0, 0.25, 0.5\}$, $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks, $\lambda_{cor} = \ln 2/t_{1/2}$, and `nlme::lme` with `corCAR1` analysis. 1,000 replicates per cell (40 minutes, 9 cores).

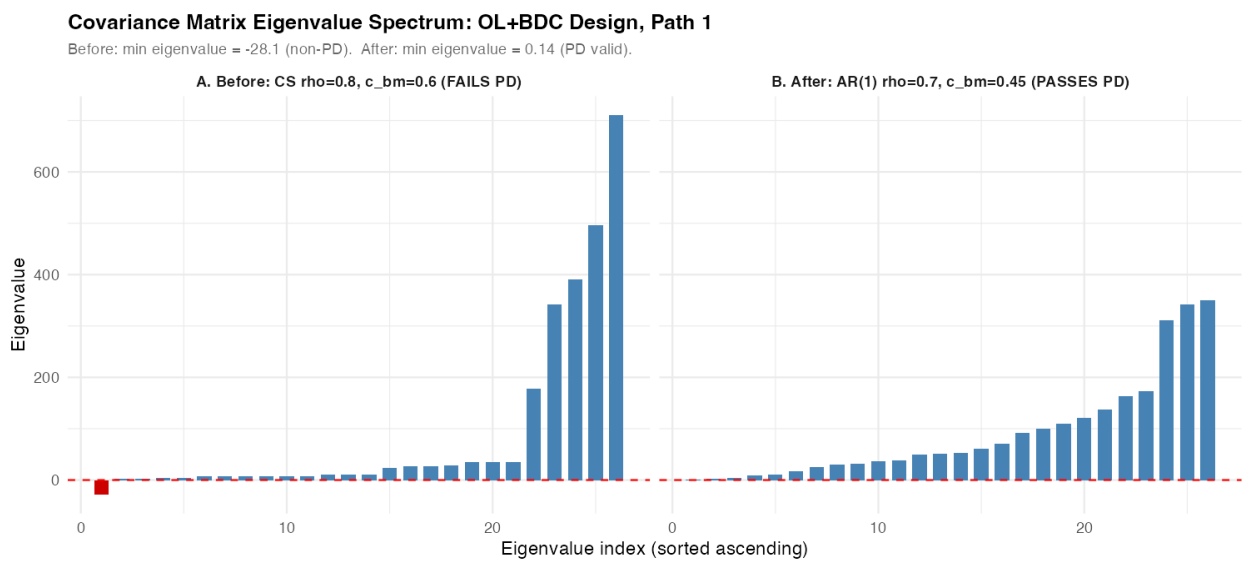

Figure 3: Eigenvalue spectrum of the covariance matrix for OL+BDC Path 1. **A:** Compound symmetry ($\rho = 0.8$, $c_{bm} = 0.6$): one eigenvalue is -28.1, rendering the matrix non-positive definite. **B:** AR(1) ($\rho = 0.7$, $c_{bm} = 0.45$): all eigenvalues positive (minimum $= 0.14$). The red dashed line marks zero.

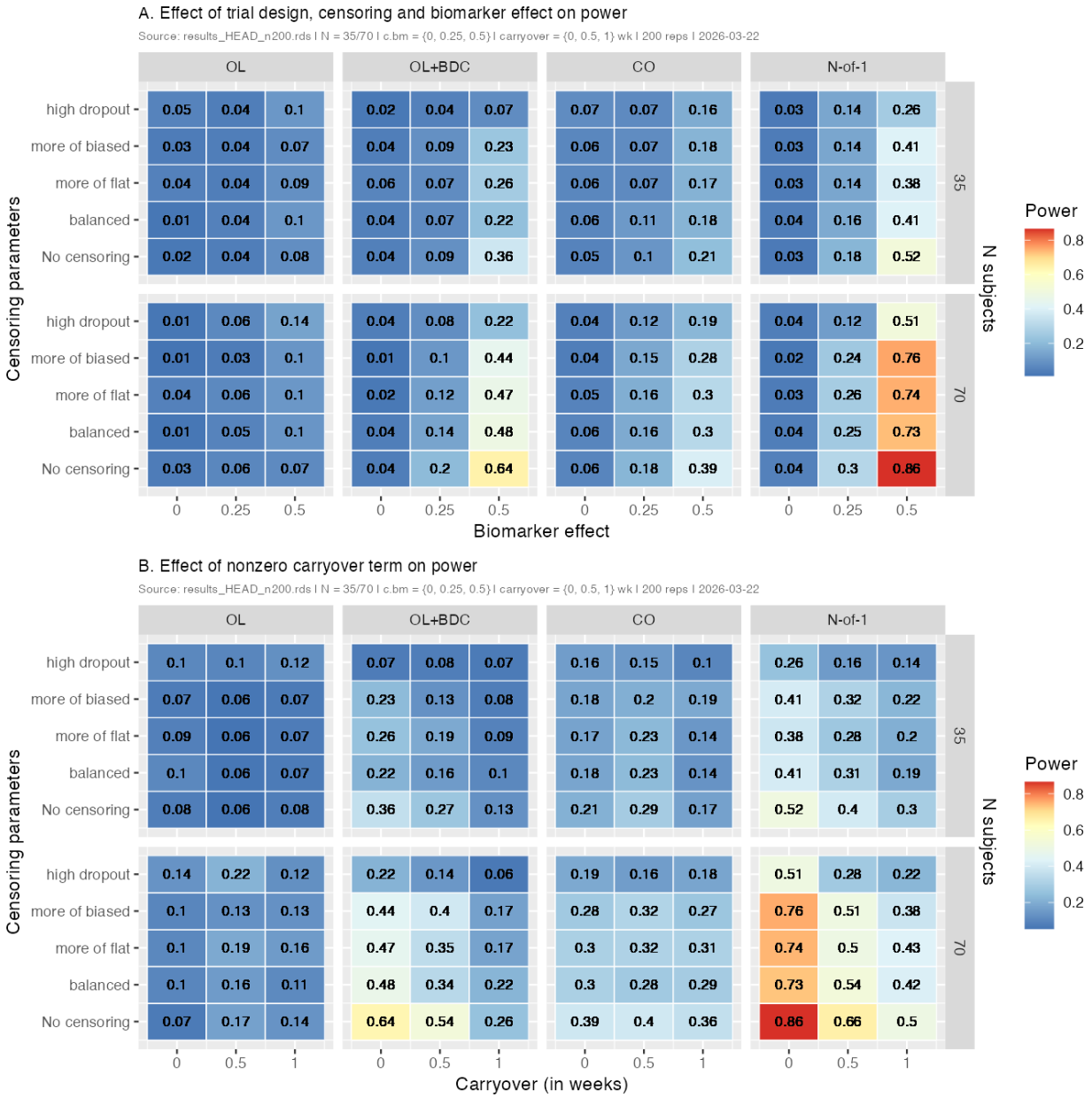

Figure 4: Figure 4 from the optimized 200-replicate simulation (8 minutes, 9 cores). Compare with Figure 2 (non-optimized, 100 minutes). The power estimates, Type I error rates, and carryover sensitivity patterns are statistically equivalent, confirming that the optimizations are result-neutral.